Repository Appendix E: Extended Interpretation and Limitations for "Benchmark-Gated Composite Mean Estimation: Regime-Conditional Improvement and Its No-Free-Lunch Limits"

Alvaro Rivera-Eraso[a,d,*], Fernando Marmolejo-Ramos[b], Freddy Hernandez-Barajas[c], Debarati Chakraborty[a], Philomena Marfo[e], Raydonal Ospina[f]

[a]*School of Computer Science, University of Hull, Hull, United Kingdom*
[b]*College of Education, Psychology and Social Work, Flinders University, Adelaide, Australia*
[c]*Departamento de Estadística, Universidad Nacional de Colombia, Medellín, Colombia*
[d]*Fundación Universitaria Konrad Lorenz, Bogotá, Colombia*
[e]*Department of Statistics and Actuarial Science, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana*
[f]*Departamento de Estatística, Universidade Federal da Bahia, Salvador, BA, Brazil*

Relationship to the journal submission

This repository appendix accompanies the article and preserves the extended interpretation, deployment discussion, limitations, and future research agenda. The journal supplementary file retains Appendices A–C, while detailed result narratives are provided in Repository Appendix D. The version-controlled project is available at `https://github.com/ALRIER/Genetic_Algo_Hull_University`. This document is interpretive support and does not replace the evidence or methods reported in the article and journal supplementary material.

*Corresponding author.
Email address: `alvaro.rivera.eraso@gmail.com` (Alvaro Rivera-Eraso)

Repository Appendix E: Extended interpretation, deployment, and limitations

E.1 Extended discussion

The main theoretical question in this study is not whether evolutionary search can generate a composite estimator, but under which conditions a weighted estimator can improve on strong classical and modern robust benchmarks. The results suggest that such improvement becomes possible when the distribution creates enough variance, tail instability, or contamination sensitivity for the sample mean and individual robust procedures to leave an exploitable region between efficiency and robustness. The genetic algorithm can search for that region, but it cannot create it. A composite succeeds only when the reduction in variance and extreme loss is large enough to compensate for the finite-sample bias introduced by components that do not individually target the arithmetic mean.

This interpretation follows directly from the MSE dominance condition developed in Section 2 of the main text. For a fixed weight vector w, improvement requires

$$B_n(w, F)^2 + V_n(w, F) < \mathrm{Var}_F(\bar{X}_n).$$

The inequality explains both the successful and unsuccessful cases. Under right skewness, heavy tails, and contamination, the variance and upper-tail loss of the sample mean may become sufficiently large for a weighted combination to accept a limited increase in bias and still reduce total MSE. This is consistent with the classical robustness literature, which shows that small amounts of contamination can greatly affect moment-based estimators, and with modern work showing that robust mean estimation depends on assumptions about tails, variance, and contamination structure (Huber, 1964; Huber and Ronchetti, 2009; Hampel et al., 1986; Wilcox, 2012; Maronna et al., 2019; Catoni, 2012; Devroye et al., 2016; Lugosi and Mendelson, 2019).

The confirmed Lognormal results fit this mechanism particularly well. Lognormal distributions combine positive support, substantial right skewness, and multiplicative dispersion (Aitchison and Brown, 1957). These properties can produce a long upper tail in the sampling distribution of the mean, especially after upper-tail contamination. In such settings, robust and partially robust components can reduce the contribution of extreme observations, while a remaining mean component helps keep the composite aligned with μ_F. The confirmed Lognormal specialists are thus consistent with a

regime in which the variance reduction obtained from trimming, bounded influence estimation, or median-based components exceeds the squared bias introduced by those same components.

Statistical confirmation and practical significance are separate decisions. The benchmark gate used in this study asks whether the frozen composite improves on both criteria, so its evidential threshold is zero gain. No universal minimum deployment gain was prespecified. A practical deployment policy should add an application-specific utility gate, consistent with selective prediction and deferral under uncertainty (Geifman and El-Yaniv, 2017; Mozannar and Sontag, 2020):

$$\underline{G}_{\text{mean},k} > \tau_{\text{mean}}, \qquad \underline{G}_{q_{0.95},k} > \tau_{q_{0.95}}, \qquad \widehat{U}_k > 0,$$

where the lower confidence bounds $\underline{G}$ must exceed thresholds fixed before deployment and $\widehat{U}_k$ incorporates implementation, monitoring, and misclassification costs. Under this distinction, CV-010 is a statistically confirmed but operationally modest specialist. Its 2.2% $q_{0.95}$ gain establishes reproducible improvement under the study design, but it does not by itself justify the added complexity of profile matching and composite computation in every application.

This result also clarifies why target compatibility matters. The median, mode, geometric mean, and harmonic mean do not generally estimate the arithmetic mean under asymmetric distributions. Their contribution can still be useful in a finite-sample composite, but only when their stabilizing effect is large enough to improve MSE for the fixed target. Gneiting's account of elicitable point functionals emphasizes that evaluation must remain tied to the stated target (Gneiting, 2011). In the present study, a component was not rewarded merely for being stable or robust. It contributed value only when the resulting composite estimated μ_F more accurately than the strongest admissible benchmark.

The same argument explains why large discovery gains did not always survive confirmation. Several Inverse Gaussian, Weibull, and Ex Wald candidates improved on the original classical benchmark set, but the margin disappeared after Catoni-type, median-of-means, winsorized, and tuned Huber estimators entered the comparison. These procedures already address many of the weaknesses of the sample mean under heavy tails and contamination (Catoni, 2012; Devroye et al., 2016; Lugosi and Mendelson, 2019, 2021; Minsker, 2023; Wilcox, 2012). Once they became available as competitors, the residual room for improvement became much smaller.

This finding supports an important condition for composite success: an estimator produced by the genetic algorithm can outperform a benchmark only when its combined influence profile captures useful structure that the benchmark does not already exploit. A mixture has no automatic advantage over a strong single estimator. When Catoni-type, median-of-means, tuned Huber, or winsorized estimators already provide an effective balance between bias and tail control, an additional convex combination may offer little benefit. The reduced accepted set under modern validation is thus evidence that the benchmark gate imposed a meaningful statistical standard.

The negative control results reinforce this interpretation. Under clean or near-efficient Normal settings, the sample mean has little excess variance available for a composite to remove. In that region, adding robust components can introduce bias or efficiency loss without generating enough protection to compensate. This matches the expectation from classical estimation theory that the sample mean is difficult to improve under Gaussian squared error conditions (Lehmann and Casella, 1998; van der Vaart, 1998). The continued retention of the benchmark in Normal regimes is not an incidental outcome. It is evidence that the dominance region of a nontrivial composite contracts when F approaches a distribution under which the mean is already efficient.

The Ex Gaussian results show that skewness alone is not sufficient. Ex Gaussian distributions can be asymmetric and long tailed, yet no transferable composite emerged in the simulation controls. The benefit of weighting depends on more than a broad family label. It depends on the complete regime: contamination mechanism, contamination rate, outlier scale, sample size, and the relationship between these factors and the influence profiles of the available components. A family can be difficult in general and still offer no stable composite advantage once a strong benchmark is available. This finding argues against treating skewness or heavy tails as sufficient activation rules.

The expanded-basis experiment directly tested another theoretical expectation: the set of attainable composites depends on the estimator library. Research on aggregation, model averaging, stacking, and the Super Learner shows that a learned combination is constrained by the candidate procedures available to it (Yang, 2001; van der Laan et al., 2007; Claeskens and Hjort, 2008; Hoeting et al., 1999). The same principle appears in evolutionary computation, where representation defines the solutions that a search procedure can express (Bäck, 1996; Eiben and Smith, 2015; Telikani et al., 2021; Katoch

et al., 2021). A genetic algorithm cannot assign weight to an estimator that is absent from its chromosome.

The shift from Lognormal specialists under the ten-component basis to Weibull specialists under the 26-component basis is consistent with this principle. Adding Catoni-type, median-of-means, winsorized, and tuned Huber components did not simply increase the number of available coordinates. It expanded the set of possible influence profiles, bias patterns, and covariance structures that could be formed inside the simplex. The new Weibull specialists could combine modern robust components with classical summaries in ways that were mathematically unavailable to the initial search.

This does not imply that a larger basis must always perform better. A richer library also creates a larger search space, increases the number of weak or redundant combinations, and can make useful regions more difficult to locate. Aggregation theory does not guarantee that every expansion of the library improves realized performance. It states that the achievable solution depends on what the library contains and on how effectively the learning rule navigates it (Yang, 2001; van der Laan et al., 2007; Claeskens and Hjort, 2008). The expanded stage supports the first part of this claim because the specialist map changed. The Dirichlet audit also speaks to the second part because it identified useful regions that the completed genetic algorithm run had not reached.

The Weibull specialists reveal another central condition: estimator performance depends on the full data-generating regime, not only on distributional family. Their weights produced stable gains under locked-unseen regimes with extreme contamination but did not retain an advantage under the original-regime evaluation. In the language of the MSE decomposition, the same vector w encountered different values of $B_n(w, F)$ and $V_n(w, F)$ after the regime changed. A weighting structure that reduced extreme variance enough to dominate under one contamination profile could retain too much bias, or achieve too little variance reduction, under another.

This result is not evidence of a general Weibull estimator. It identifies a narrower dominance region defined by Weibull structure together with severe upper-tail or point-mass contamination. The family label alone is insufficient for deployment. Activation must depend on a more detailed profile that includes the contamination pattern and scale. This is one of the most important findings of the study because it moves the idea of regime dependence from a broad conceptual claim to an operational restriction.

The asymmetry between matched benefit and mismatched loss requires

a conservative pre-activation rule. The benchmark remains the default. A specialist may be considered only when an independent diagnostic, specified without using the candidate's observed performance, supports the validated family, contamination direction and structure, severity range, and sample-size range. Activation must also clear the practical utility gate above. If the profile is ambiguous, falls near a boundary, changes during use, or lacks a validated diagnostic mapping, the procedure abstains and retains the benchmark, following the established logic of reject-option and learning-to-defer systems (Geifman and El-Yaniv, 2017; Mozannar and Sontag, 2020). One possible utility representation is

$$\widehat{U}_k = (1 - \widehat{p}_{\mathrm{mis}})\widehat{G}_k - \widehat{p}_{\mathrm{mis}}\widehat{L}_k - C_k,$$

where $\widehat{p}_{\mathrm{mis}}$ is the estimated probability of profile misclassification, $\widehat{G}_k$ is the expected matched-regime benefit, $\widehat{L}_k$ is the expected mismatched-regime loss, and C_k is the implementation cost. The current study does not estimate $\widehat{p}_{\mathrm{mis}}$ or validate an autonomous real-time classifier. The specialists are therefore validated statistical configurations, not autonomous deployment rules. The large negative original-regime outcomes and the full candidate matrix in Supplementary Table S3 in the journal supplementary material make this restriction explicit.

The difference between discovery in one regime and confirmed gain in a related locked-unseen regime is also informative. Evolutionary discovery can identify a region of the simplex whose usefulness extends beyond the exact cells that produced it, but the location of its strongest advantage may shift within the surrounding structural neighborhood. In this sense, discovery does not necessarily recover the single best vector for one fixed cell. It can uncover a weighting pattern associated with a broader local geometry of related regimes. Confirmation then identifies where within that geometry the frozen vector actually satisfies the dominance condition.

The robustness audit helps explain how the confirmed Weibull composites achieved this behavior. Both specialists retained positive weight on the sample mean, so their formal breakdown point remains zero. Classical influence-function theory makes clear that an unbounded component with positive weight transfers part of its unbounded influence to the composite (Hampel et al., 1986; Huber and Ronchetti, 2009; Avella-Medina, 2017). Yet the magnitude of that contribution was strongly reduced. The estimated effect of one arbitrarily large observation was approximately 1/62 and 1/46 of its effect on the sample mean.

This distinction matters. A breakdown point of zero does not mean that the composite behaves identically to the sample mean under practically relevant contamination. Formal breakdown describes what can happen under arbitrarily severe replacement contamination. Gross error sensitivity describes how strongly a small contamination perturbation can affect the estimator (Hampel et al., 1986; Huber and Ronchetti, 2009). The two specialists did not acquire full high-breakdown robustness, but they substantially attenuated the influence of extreme observations while preserving some alignment with the arithmetic mean. Their gains are consistent with this intermediate position between efficiency and robustness.

The composition of the weights supports the same interpretation. Most of the mass was assigned to bounded or controlled influence procedures, including Huber, Catoni-type, biweight, median, and winsorized estimators. Modern work on M estimators and robust means shows how bounded influence and deviation control can stabilize estimation under contamination and heavy tails (Mathieu, 2022; Catoni, 2012; Wang and Ramdas, 2023; Lugosi and Mendelson, 2019). The small positive mean weight prevented complete separation from the target, while the robust components reduced exposure to the upper tail. The resulting estimator was not maximally robust in a formal sense; it was tuned to reduce risk for a specific regime.

The confirmation audits also address whether the observed dominance could be explained by random seed selection or post-selection optimism. The agreement between pooled and split-sample estimates indicates that the gains did not depend materially on using the same complete validation collection for selection and reporting. The survival of the specialists across eight validation seeds and under false discovery rate correction further supports the stability of the signal (Benjamini and Hochberg, 1995). These checks do not create the advantage, but they reduce the plausibility that it arose from an isolated Monte Carlo realization.

The bootstrap audit introduces a narrower caveat. Ordinary bootstrap intervals can behave poorly for medians and mode-based procedures (Ghosh et al., 1984; Romano, 1988). Modal weight was negligible in the confirmed Weibull specialists, which limits its practical impact on the reported intervals. Median weight was more visible in one specialist, so the finite-sample approximation may still converge more slowly than it would for a fully smooth estimator. This issue does not overturn the current result, but it identifies a specific extension: future confirmation can compare the ordinary paired bootstrap with m out of n subsampling when nonsmooth components re-

ceive substantial weight.

The random Dirichlet audit also changes how abstention should be interpreted. In many cells, 4,000 independent simplex draws failed to produce a vector that passed both MSE gates across all seeds. Those results indicate that no stable gate-passing configuration was found under the tested basis, Dirichlet proposal distribution, and audit budget. The recovery of both Weibull specialists as positive controls, together with continued abstention in all Normal and Ex Gaussian controls, shows that the audit could distinguish known valid regions from benchmark-dominant settings.

At the same time, the audit found gate-passing configurations that remained stable across all seeds in eight regimes not recovered by the completed genetic algorithm run. The strongest case, Q1R003, had a pass rate of 1.8%, while the remaining rates were below 0.3%. These frequencies indicate that successful configurations can be uncommon under the selected Dirichlet proposal distribution. They do not by themselves establish the volume, connectivity, or geometry of the corresponding sets within the simplex.

The result also shows that benchmark retention can have two different interpretations. In some regimes, the audit found no stable passing configuration within its sampling budget. In others, passing configurations were present but were not recovered by the completed genetic algorithm run. Benchmark retention may thus reflect either limited evidence of an accessible gain under the audit distribution or incomplete exploration by the search procedure.

This distinction limits what can be concluded about the genetic algorithm itself. The validated specialists remain valid because their status depends on frozen evaluation, not on the claim that the GA is the best optimizer. The Dirichlet results suggest that alternative mutation schedules, larger populations, longer persistence, CMA-ES, mirror descent, or other simplex methods could improve discovery efficiency (Beck and Teboulle, 2003; Hansen, 2016; Bergstra and Bengio, 2012). Such comparisons would address how effectively the search finds candidate regions. They would not change the benchmark gate that determines whether a persisted vector deserves confirmation.

The external evaluation provides a final test of the regime matching hypothesis. The frozen specialists achieved corrected wins across multiple independent parent datasets and repeated subregimes without retraining. These results indicate that some weighting patterns learned in synthetic regimes can remain useful when applied to empirical distributions with compatible profiles. The use of independent parent datasets as the main unit of evidence

also avoids counting nested evaluations as independent replications (Hurlbert, 1984; Aarts et al., 2014).

The external signal was selective. It appeared in Lognormal, Normal, and Weibull profiles and was absent from Ex Gaussian, Ex Wald, and Inverse Gaussian profile-eligible evaluations. This pattern confirms that simulation success does not map mechanically onto every empirical family. It also shows that a specialist discovered under one synthetic condition may contribute to an empirical profile only when the observed distribution, contamination pattern, and benchmark relationship are sufficiently compatible.

The appearance of external wins in Normal profiles deserves careful interpretation because Normal regimes served as negative controls during simulation. The result does not necessarily indicate a contradiction. The external profile classification captures empirical similarity, not exact Gaussian sampling under the clean simulation condition. A dataset assigned to a Normal profile may still contain finite-sample irregularities, endogenous contamination, heterogeneity, or tail behavior that creates room for a specialist to improve recovery of the full-sample reference mean. The correct conclusion is that the activation rule must be based on the complete empirical profile, not the family label in isolation.

The external target also limits the claim. The full-sample mean served as the reference because the true population mean is unknown in public observational data. The external analysis thus evaluates recovery of an empirical reference from repeated subsamples. It does not establish that the same improvement would hold against an independently known population parameter. Even within this restricted target, the consistency between simulation and external activation supports the underlying claim that estimator performance depends on matching the weighting structure to the regime.

The findings align with the structural lesson of the No Free Lunch literature, although they do not constitute a proof of the formal optimization theorem. Wolpert and Macready show that no optimizer dominates when performance is averaged across unrestricted problem classes, while later work emphasizes the role of inductive biases and restrictions on the environment (Wolpert and Macready, 1997; Wolpert, 2023; Sterkenburg and Grünwald, 2021; Goldblum et al., 2024). The present study expresses the same logic in estimator selection: weights that are useful under one combination of skewness, tail behavior, and contamination can lose their advantage when those conditions change.

The results also show that conditional performance is not equivalent to

arbitrary post hoc specialization. The specialist configurations were subjected to frozen validation, new seeds, locked-unseen conditions, multiplicity control, negative controls, and external testing. These safeguards separate a genuine local dominance region from a candidate that only fits the discovery simulations. The benchmark gate is central because it permits abstention when the evidence does not support activation.

The broader methodological implication is that the GA should be understood as a generator of hypotheses over the estimator simplex. Statistical authority comes from the later comparison with admissible benchmarks. Under favorable regimes, evolutionary search can combine complementary influence profiles and identify a vector that reduces both average and upper-tail loss. When the mean is near-efficient, the benchmark already captures the available gain, or the regime is mismatched, the same process should return no usable composite.

The evidence supports a clear set of conditions for success. A weighted estimator produced by the genetic algorithm is most likely to improve on the benchmark when the sample mean has substantial variance driven by the upper tail, no single robust benchmark fully captures the regime structure, the estimator basis can express a suitable balance between bias and influence, and the learned vector is applied within a compatible validation region. It is unlikely to improve when the mean is already efficient, when a modern robust benchmark absorbs most of the available gain, when the relevant component is missing from the basis, or when the deployment regime changes the balance between bias and variance.

In this sense, the main contribution is not a claim that evolutionary weighting produces a superior estimator in general. It is a tested account of when such weighting can be useful. The search identifies possible regions of advantage, the benchmark gate determines whether the gain is real, and the profile rule limits application to conditions supported by the evidence. Retaining the benchmark outside those regions is part of the intended behavior of the method, not a failure of discovery.

E.2 Extended limitations and future research

The first limitation concerns the range of statistical environments represented in the study. The simulation design assumes independent and identically distributed observations, is univariate, and focuses on sample sizes $n \geq 300$. This scope made it possible to isolate how distributional

shape, contamination mechanism, severity, and estimator composition affect the recovery of μ_F. It does not yet cover dependence between observations, multivariate targets, relatively small samples, censoring, or changes in the data-generating process over time. These extensions are especially relevant for reaction time research, where trials are nested within participants, individual distributions can differ substantially, and temporal effects such as learning, fatigue, and adaptation may violate simple iid assumptions (Luce, 1986; Brenner and Smeets, 2019; Tejo et al., 2018; Chávez De la Peña and Vandekerckhove, 2025). Future work should examine hierarchical and participant-specific specialist maps, with validation performed at both trial and participant levels. Such designs would test whether a simplex region learned from pooled observations remains useful after within-subject dependence and between-subject heterogeneity are represented explicitly.

The public data battery provides an important extension beyond simulation, but its target remains operational. Because the population mean is unknown in observational datasets, the full-sample mean was used as the reference for repeated subsample recovery. The external findings show that frozen specialists can recover this empirical target more accurately under matched profiles, but they do not establish performance against an independently known population parameter. Future external studies could use very large reference samples, administrative populations, controlled experimental archives, or prospective data collection to create a stronger approximation to the true target. In reaction time applications, this could include high-trial-count experiments in which a large participant-level distribution serves as the reference and smaller trial sets reproduce the practical conditions faced by applied researchers.

The empirical contamination rule also deserves further study. Public data irregularities were detected through the fixed $z = 3$ rule based on the median and unscaled MAD, with no synthetic contamination added. This choice supports a reproducible external evaluation, but the present study does not include a sensitivity analysis of the threshold. In right-skewed Lognormal or Weibull marginals, legitimate tail observations may be flagged as operational irregularities even when they arise from the main distribution (Berger and Kiefer, 2021; Miller, 2023). The external results are consequently conditional on $z = 3$ and should not be interpreted as invariant to the diagnostic threshold. A focused extension should repeat the frozen external battery at $z \in \{2.5, 3, 4\}$ and compare eligible parents, profile-matched comparisons, CI-confirmed wins, survivors after false discovery rate correction, and family-

specific activation counts. Reaction time observations may contain fast guesses, attentional lapses, deadline effects, or participant-specific secondary processes, each of which can produce a different tail structure (Schwarz, 2001; Heathcote, 2004; Marmolejo-Ramos et al., 2023).

A second limitation concerns the discovery map. Candidate nomination rests on two discovery seeds, while confirmation uses eight additional seeds, higher Monte Carlo precision, paired bootstrap intervals, locked-unseen regimes, and multiplicity control. The confirmed claims do not depend solely on the two discovery streams, but the number and location of nominated candidates may still vary under a larger set of discovery seeds, a form of selection variability that can generate optimistic candidate estimates when discovery and assessment are not adequately separated (Cawley and Talbot, 2010; van Zwet and Cator, 2021). A natural extension is to estimate selection probabilities for each region of the simplex. Repeated discovery runs could show whether the same weight structures appear consistently, whether several equivalent solutions occupy a common neighborhood, or whether a candidate depends on an unusually favorable evolutionary trajectory. This would refine the map of discoverable regions without changing the frozen validation standard used to evaluate any persisted candidate.

The optimizer itself also remains open to comparison. The study uses a genetic algorithm because the objective is stochastic, nonsmooth, and constrained to the simplex, and because population-based search supports warm starts, mutation, and persistence across related regimes. The findings do not establish that the GA is the most efficient method for locating gate-passing regions. The Dirichlet audit showed that some useful areas may occupy very small portions of the simplex and may not be reached by a completed evolutionary run. Equal-budget comparisons with random search, CMA-ES, projected or mirror descent, Bayesian optimization, and alternative evolutionary designs would help determine which search strategy identifies these regions most efficiently (Bergstra and Bengio, 2012; Beck and Teboulle, 2003; Hansen, 2016; Bäck, 1996; Eiben and Smith, 2015). This comparison concerns discovery efficiency. The evidential status of a candidate would still depend on frozen weights, benchmark comparison, and post-discovery confirmation.

The expanded-basis results also motivate a more systematic study of estimator libraries. Adding Catoni-type, median-of-means, winsorized, and tuned Huber components changed which specialist regions could be expressed and produced the confirmed Weibull candidates. A larger library can create

new influence and bias combinations, but it can also introduce redundant coordinates and increase search difficulty (van der Laan et al., 2007; Claeskens and Hjort, 2008; Yang, 2001). Future studies should compare nested estimator bases and measure the marginal value of each component family. Such an analysis could determine whether a specialist depends on a particular modern robust estimator, on covariance between several components, or simply on the greater flexibility of the expanded simplex. Prespecified ablations of warm starts, inheritance, mutation concentration, and HPF exposure would complement this analysis.

Multiplicity control is concentrated at the stages where evidential claims are made. Discovery serves as a screening process and does not independently establish significance. Fixed-weight confirmation uses paired uncertainty estimates and a Benjamini-Hochberg correction across confirmation candidates, while the external battery applies false discovery rate control across profile-eligible comparisons (Benjamini and Hochberg, 1995). The correction reported in Section 4.4 of the main text left the confirmed set unchanged. Future studies could strengthen this design through a fully preregistered confirmation list, independent simulation batches reserved before discovery, or sequential error control when candidates are evaluated over several waves (Robertson et al., 2023). These changes would provide an even sharper separation between exploratory search and confirmatory evidence.

The evidence taxonomy should also be developed prospectively. The internal categories summarize existing fixed-weight validation outcomes, but their detailed definitions were finalized after the broad simulation patterns were known. They are frozen and versioned for transparent reporting, yet they are not treated as retrospectively preregistered (Nosek et al., 2018). The external eligibility, benchmark policy, and false discovery procedure were specified before external wins were counted. Future work should apply the same prospective standard to the internal taxonomy, including explicit criteria for transfer specialists, local candidates, near-gate cases, and negative controls.

The Dirichlet audit introduces a further distinction that future studies should investigate directly. Benchmark retention can indicate that no stable gain exists in the evaluated regime, or that a useful region exists but occupies such a small part of the simplex that the completed search did not locate it. Persisting the strongest Dirichlet vectors and passing them through the full frozen validation procedure would test these two explanations. Q1R003 and the other flagged regimes offer concrete starting points.

This extension could also examine the local structure surrounding confirmed and audit-identified configurations. Systematic perturbations of the weights could determine whether successful vectors belong to broad neighborhoods of similar performance or to more localized sets requiring targeted search. The present Dirichlet audit identifies candidate configurations but does not estimate their geometry or volume within the simplex.

The Catoni-type components use a fixed implementation-specific sensitivity grid instead of the variance- and confidence-dependent calibration required for Catoni's original deviation guarantee. Their contribution should therefore be interpreted as empirically validated within the scales and regimes examined here, not as an implementation inheriting Catoni's theoretical finite-sample bound. Scale-adaptive or plug-in calibration is an important extension for future work.

Bibliography

Aarts, E., Verhage, M., Veenvliet, J.V., Dolan, C.V., van der Sluis, S., 2014. A solution to dependency: Using multilevel analysis to accommodate nested data. Nature Neuroscience 17, 491–496. doi:10.1038/nn.3648.

Aitchison, J., Brown, J.A.C., 1957. The Lognormal Distribution. Cambridge University Press, Cambridge. With special reference to its uses in economics.

Avella-Medina, M., 2017. Influence functions for penalized M-estimators. Bernoulli 23, 3178–3196. doi:10.3150/16-BEJ841.

Bäck, T., 1996. Evolutionary Algorithms in Theory and Practice: Evolution Strategies, Evolutionary Programming, Genetic Algorithms. Oxford University Press, New York, NY. doi:10.1093/oso/9780195099713.001.0001.

Beck, A., Teboulle, M., 2003. Mirror descent and nonlinear projected subgradient methods for convex optimization. Operations Research Letters 31, 167–175. doi:10.1016/S0167-6375(02)00231-6.

Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. Journal of the Royal Statistical Society: Series B (Methodological) 57, 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.

Berger, A., Kiefer, M., 2021. Comparison of different response time outlier exclusion methods: A simulation study. Frontiers in Psychology 12, 675558. doi:10.3389/fpsyg.2021.675558.

Bergstra, J., Bengio, Y., 2012. Random search for hyper-parameter optimization. Journal of Machine Learning Research 13, 281–305. URL: https://www.jmlr.org/papers/v13/bergstra12a.html.

Brenner, E., Smeets, J.B.J., 2019. How can you best measure reaction times? Journal of Motor Behavior 51, 486–495. doi:10.1080/00222895.2018.1518311.

Catoni, O., 2012. Challenging the empirical mean and empirical variance: A deviation study. Annales de l'Institut Henri Poincaré—Probabilités et Statistiques 48, 1148–1185. doi:10.1214/11-AIHP454.

Cawley, G.C., Talbot, N.L.C., 2010. On over-fitting in model selection and subsequent selection bias in performance evaluation. Journal of Machine Learning Research 11, 2079–2107. URL: https://www.jmlr.org/papers/v11/cawley10a.html.

Claeskens, G., Hjort, N.L., 2008. Model Selection and Model Averaging. Cambridge University Press, Cambridge. doi:10.1017/CBO9780511790485.

Devroye, L., Lerasle, M., Lugosi, G., Oliveira, R.I., 2016. Sub-Gaussian mean estimators. The Annals of Statistics 44, 2695–2725. doi:10.1214/16-AOS1440.

Eiben, A.E., Smith, J.E., 2015. Introduction to Evolutionary Computing. Natural Computing Series. 2 ed., Springer, Berlin, Heidelberg. doi:10.1007/978-3-662-44874-8.

Geifman, Y., El-Yaniv, R., 2017. Selective classification for deep neural networks, in: Advances in Neural Information Processing Systems 30, pp. 4885–4894. URL: https://proceedings.neurips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html.

Ghosh, M., Parr, W.C., Singh, K., Babu, G.J., 1984. A note on bootstrapping the sample median. The Annals of Statistics 12, 1130–1135. doi:10.1214/aos/1176346731.

Gneiting, T., 2011. Making and evaluating point forecasts. Journal of the American Statistical Association 106, 746–762. doi:10.1198/jasa.2011.r10138.

Goldblum, M., Finzi, M.A., Rowan, K., Wilson, A.G., 2024. Position: The no free lunch theorem, Kolmogorov complexity, and the role of inductive biases in machine learning, in: Proceedings of the 41st International Conference on Machine Learning, PMLR. pp. 15788–15808. URL: https://proceedings.mlr.press/v235/goldblum24a.html.

Hampel, F.R., Ronchetti, E.M., Rousseeuw, P.J., Stahel, W.A., 1986. Robust Statistics: The Approach Based on Influence Functions. John Wiley & Sons, New York. doi:10.1002/9781118186435.

Hansen, N., 2016. The CMA evolution strategy: A tutorial. URL: https://arxiv.org/abs/1604.00772, doi:10.48550/arXiv.1604.00772, arXiv:1604.00772. arXiv preprint.

Heathcote, A., 2004. Fitting Wald and ex-Wald distributions to response time data. Behavior Research Methods, Instruments, & Computers 36, 678–694. doi:10.3758/BF03195574.

Hoeting, J.A., Madigan, D., Raftery, A.E., Volinsky, C.T., 1999. Bayesian model averaging: A tutorial. Statistical Science 14, 382–401. doi:10.1214/ss/1009212519.

Huber, P.J., 1964. Robust estimation of a location parameter. The Annals of Mathematical Statistics 35, 73–101. doi:10.1214/aoms/1177703732.

Huber, P.J., Ronchetti, E.M., 2009. Robust Statistics. Wiley Series in Probability and Statistics. 2 ed., John Wiley & Sons, Hoboken, NJ. doi:10.1002/9780470434697.

Hurlbert, S.H., 1984. Pseudoreplication and the design of ecological field experiments. Ecological Monographs 54, 187–211. doi:10.2307/1942661.

Katoch, S., Chauhan, S.S., Kumar, V., 2021. A review on genetic algorithm: Past, present, and future. Multimedia Tools and Applications 80, 8091–8126. doi:10.1007/s11042-020-10139-6.

Lehmann, E.L., Casella, G., 1998. Theory of Point Estimation. Springer Texts in Statistics. 2 ed., Springer, New York, NY. doi:10.1007/b98854.

Luce, R.D., 1986. Response Times: Their Role in Inferring Elementary Mental Organization. Oxford University Press, New York.

Lugosi, G., Mendelson, S., 2019. Mean estimation and regression under heavy-tailed distributions: A survey. Foundations of Computational Mathematics 19, 1145–1190. doi:10.1007/s10208-019-09427-x.

Lugosi, G., Mendelson, S., 2021. Robust multivariate mean estimation: The optimality of trimmed mean. The Annals of Statistics 49, 393–410. doi:10.1214/20-AOS1961.

Marmolejo-Ramos, F., Barrera-Causil, C., Kuang, S., Fazlali, Z., Wegener, D., Kneib, T., De Bastiani, F., Martinez-Flórez, G., 2023. Generalised exponential-Gaussian distribution: A method for neural reaction time analysis. Cognitive Neurodynamics 17, 221–237. doi:10.1007/s11571-022-09813-2.

Maronna, R.A., Martin, R.D., Yohai, V.J., Salibián-Barrera, M., 2019. Robust Statistics: Theory and Methods (with R). Wiley Series in Probability and Statistics. 2 ed., John Wiley & Sons, Chichester. doi:10.1002/9781119214656.

Mathieu, T., 2022. Concentration study of M-estimators using the influence function. Electronic Journal of Statistics 16, 3695–3750. doi:10.1214/22-EJS2030.

Miller, J., 2023. Outlier exclusion procedures for reaction time analysis: The cures are generally worse than the disease. Journal of Experimental Psychology: General 152, 3189–3217. doi:10.1037/xge0001450.

Minsker, S., 2023. Efficient median of means estimator, in: Neu, G., Rosasco, L. (Eds.), Proceedings of the Thirty-Sixth Conference on Learning Theory, PMLR. pp. 5925–5933. URL: https://proceedings.mlr.press/v195/minsker23a.html.

Mozannar, H., Sontag, D., 2020. Consistent estimators for learning to defer to an expert, in: Proceedings of the 37th International Conference on Machine

Learning, pp. 7076–7087. URL: https://proceedings.mlr.press/v119/mozannar20b.html.

Nosek, B.A., Ebersole, C.R., DeHaven, A.C., Mellor, D.T., 2018. The preregistration revolution. Proceedings of the National Academy of Sciences 115, 2600–2606. doi:10.1073/pnas.1708274114.

Chávez De la Peña, A.F., Vandekerckhove, J., 2025. An EZ bayesian hierarchical drift diffusion model for response time and accuracy. Psychonomic Bulletin & Review 32, 3067–3087. doi:10.3758/s13423-025-02729-y.

Robertson, D.S., Wason, J.M.S., Ramdas, A., 2023. Online multiple hypothesis testing. Statistical Science 38, 557–575. doi:10.1214/23-STS901.

Romano, J.P., 1988. Bootstrapping the mode. Annals of the Institute of Statistical Mathematics 40, 565–586. doi:10.1007/BF00053066.

Schwarz, W., 2001. The ex-Wald distribution as a descriptive model of response times. Behavior Research Methods, Instruments, & Computers 33, 457–469. doi:10.3758/BF03195403.

Sterkenburg, T.F., Grünwald, P.D., 2021. The no-free-lunch theorems of supervised learning. Synthese 199, 9979–10015. doi:10.1007/s11229-021-03233-1.

Tejo, M., Niklitschek-Soto, S., Marmolejo-Ramos, F., 2018. Fatigue-life distributions for reaction time data. Cognitive Neurodynamics 12, 351–356. doi:10.1007/s11571-017-9473-x.

Telikani, A., Tahmassebi, A., Banzhaf, W., Gandomi, A.H., 2021. Evolutionary machine learning: A survey. ACM Computing Surveys 54, 1–35. doi:10.1145/3467477. article 161.

van der Laan, M.J., Polley, E.C., Hubbard, A.E., 2007. Super learner. Statistical Applications in Genetics and Molecular Biology 6, Article 25. doi:10.2202/1544-6115.1309.

van der Vaart, A.W., 1998. Asymptotic Statistics. Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, Cambridge. doi:10.1017/CBO9780511802256.

van Zwet, E.W., Cator, E.A., 2021. The significance filter, the winner's curse and the need to shrink. Statistica Neerlandica 75, 437–452. doi:10.1111/stan.12241.

Wang, H., Ramdas, A., 2023. Catoni-style confidence sequences for heavy-tailed mean estimation. Stochastic Processes and their Applications 163, 168–202. doi:10.1016/j.spa.2023.05.007.

Wilcox, R.R., 2012. Introduction to Robust Estimation and Hypothesis Testing. 3 ed., Academic Press, Waltham, MA.

Wolpert, D.H., 2023. The implications of the no-free-lunch theorems for meta-induction. Journal for General Philosophy of Science 54, 421–432. doi:10.1007/s10838-022-09609-2.

Wolpert, D.H., Macready, W.G., 1997. No free lunch theorems for optimization. IEEE Transactions on Evolutionary Computation 1, 67–82. doi:10.1109/4235.585893.

Yang, Y., 2001. Adaptive regression by mixing. Journal of the American Statistical Association 96, 574–588. doi:10.1198/016214501753168262.